

Training Your Models Faster and More Accurately

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Accuracy comes mostly from clean, varied labels, not from training longer; speed comes from a graphics card, the training-site tour, and retraining in batches. The site's methods are sound and close to the best that runs in a browser, but four upgrades from current research would help: a by-area accuracy check, extra terrain views, a pretrained network, and outlines instead of points.

Quick wins

Do these in order; the first two matter more than anything else on this page.

1. **Clean the mound data you have.** Open **Review training images**, sort by *Lowest first*, and fix every "mound" that shows no clear rise in the middle. In the file you sent earlier, 61 of 135 confirmed mounds were wider than 115 ft, so the network never used them, and several were recorded as 0 ft tall.
2. **Run the training data check after each training.** It flags mounds the network doubts and "Not real" marks it thinks are mounds. Sort the image review by *Network disagrees most first* to see them as pictures.
3. **Add hard negatives.** Answer *Not real* on the deep-learning scan's false alarms (tree throws, root balls, spoil piles, road crowns). Hard negatives teach the network more than random ground.
4. **Use the tour for volume.** Each toured site adds candidates and background ground for all three models in one pass.
5. **Train once per batch, not per answer.** Review a full set of candidates, then let the tour train every model at the end.
6. **Use a desktop browser with a graphics card.** Chrome or Edge on a computer with a GPU trains several times faster than a phone or a machine without one.

Clean labels

A network copies its teacher: one wrong "mound" undoes several right ones, because it learns that the wrong shape counts too.

- **Real means a clear rise centred in the picture.** If the ring in the image review sits beside the bump, use **Go to** and **Move** to put the point on the mound's top.
- **Record true height and width.** A mound saved at the size of the whole hill teaches the scans and network the wrong scale. Outline mounds whose base you can see; the outline measures both for you.
- **Keep each kind in its own model.** Effigies go in the effigy model, shell middens in the shell model, never in the mound model. The tour already drops mound candidates that sit on an effigy or midden.
- **Mark look-alikes Not real instead of deleting them.** A "Not real" mark is a lesson; a deletion is just less data. Delete only items that are mistakes in themselves (wrong place, duplicate).
- **Skip, don't guess.** In the tour review, *Not sure* costs nothing; a wrong answer costs accuracy.
- **Vary the examples.** Confirm mounds on slopes and flats, in fields and woods, conical and platform, large and small, across several regions. Thirty mounds from one park teach the network that park.
- **Check the "No image yet" filter.** Items without a terrain sample teach the network nothing until a DTM covering them is loaded.

How many examples each model needs

The minimums are hard limits built into the site; the "good" counts are my estimates, not measured on your data.

Model	Window it sees	Trains from	Held-out check from	Good results from (estimate)	Left out
Mounds	40 m square at 1 m	3 mounds	5 mounds	100+ mounds, 300+ not-mounds	Mounds wider than 35 m (115 ft)
Effigy & linear mounds	128 m square at 2 m	3 effigies	5 effigies	40+ effigies	Nothing
Shell middens & rings	192 m square at 3 m	3 middens	5 middens	40+ middens	Nothing

Background ground is collected automatically, up to 4,000 samples for mounds and 1,500 for each other model. Below about 20 real examples, treat any model's results as rough.

Training faster

The biggest speed-up is hardware; the next is not repeating work.

- **Graphics card.** Training runs on WebGL when the browser has a GPU. Without one it falls back to the processor, which is several times slower; the status line says "no graphics-card support" when that happens.
- **Desktop over phone.** Phones throttle long jobs and may pause a background tab. Keep the tab in front while training.
- **Tour site size.** *Medium (500 m)* is the default and covers most mound groups. Use *Small (300 m)* for single mounds: downloads and scans finish in roughly a third of the time.
- **Batch the work.** Tour many sites, review all candidates, then train once. Each training run uses all data from scratch, so training after every answer repeats the same work.
- **Train only what changed.** The tour's automatic training trains only the models you gave answers for.
- **Collect samples while the area is loaded.** A training item gets its terrain picture only while a DTM covering it is loaded. The tour does this for you; for items added by hand, press **Collect samples from this DTM** before moving on.

The improvement loop

Accuracy grows by repeating one short loop, not by one long training run.

Each round of answers makes the next scan better



improvement loop · 4 steps

Step 3 is the one most often skipped and the one that pays most: the network's own doubts, shown by the training data check, point straight at the labels worth fixing.

Are we using the best method?

Mostly yes for a tool that runs in a browser, but not the best available overall. The rule-based scans follow published methods, and the network training uses the standard tricks. The gaps are in how accuracy is measured, how much the network starts out knowing, and what it is shown.

Part	What the site does	Current best practice in research	Gap
Elevation data	USGS 3DEP bare-earth DEM at 1 m, or original USGS 1 m tiles	Same, or the raw LiDAR point cloud with custom ground filtering at 0.5 m	Small. Raw point clouds help most under dense brush
Rule-based mound scan	Inverted-DEM "iMound" fill, LiDAR-error persistence, dome and platform template matching	The same family of methods	None
Terrain views for the network	Relief, fine relief, steepness (mounds); narrow relief, broad relief, steepness (effigies, shells)	Several LiDAR visualizations stacked: local relief, sky-view factor, openness, slope	Medium. Sky-view factor and openness often help low features
Network	Small network trained from scratch, sliding window over the view	Detectors such as Faster R-CNN, Mask R-CNN or YOLO, usually starting from a network pretrained on millions of images	Large with few examples. Pretraining needs far fewer labels
What it predicts	"Mound here or not" for a square; size from the rule-based measurer	An outline per feature (segmentation)	Medium. Outlines give size directly and separate touching mounds
Augmentation	8 rotations and mirrors, shifts, height scaling; 8-way averaged scoring	Same	None

Part	What the site does	Current best practice in research	Gap
Negatives	Random ground plus your "Not real" marks; active learning through scan answers	Same, with hard-negative mining	None
Accuracy check	Random 1 in 5 held back	Held back by area, so test mounds are in places the network never saw	Large. A random split overstates accuracy

The research names above are from memory; I could not open the papers from here, so check details before citing them. Relevant work includes Freeland et al. (iMound), Davis, Lipo and Sanger (mound and shell-ring detection in coastal South Carolina), and Verschoof-van der Vaart and Lambers and Trier et al. (deep-learning detection of burial mounds).

What could be added next

Ranked by how much accuracy each would add for the effort. The first two can be built into the site as it is; the last two need work outside the browser.

1. **Accuracy check by area.** Hold back whole areas, not random items, so the reported accuracy is honest. Small change, and every later decision gets better evidence.
2. **Extra terrain views.** Add sky-view factor and openness to what the network sees. Both are standard in LiDAR archaeology and bring out low, broad features like plough-worn mounds. Moderate change; each model must be retrained once.
3. **A pretrained network.** Start from a network already trained on millions of images, then fine-tune it on your mounds. It typically needs far fewer labels for the same accuracy. Training would move to a computer with a GPU (Python), with the result loaded back into the site.
4. **Outlines instead of points.** A segmentation network that draws each mound's footprint. It needs outlined training mounds, which the site's outline tool already records, so start outlining now.

Until then, the quickest gains are still the first two quick wins: clean labels and more varied examples.